
Yufeng (Alex) Xia

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EDUCATION

- **University of Edinburgh** Edinburgh, UK
MSc by Research in Computer Science, supervised by Prof. Mahesh K. Marina Sep 2025 – Present
- **University of Edinburgh** Edinburgh, UK
*Bachelor of Science in Computer Science; Grade: **First Class (Hons)*** Sep 2021 – May 2025

RESEARCH EXPERIENCE

- **University of Edinburgh – Research Assistant** ICSA / NetSys Group
2024 – Present
 - **LLVM Compiler Framework:** Developed CoSenseAQ, an LLVM-based auto-quantization framework leveraging sensor specifications for code size and performance optimization on embedded ARM systems.
 - **Scalable Edge Emulator:** Participated in developing Morphling, a measurement-driven emulation platform supporting 1800+ heterogeneous edge devices for distributed ML training evaluation.
 - **AI-RAN Infrastructure:** Contributed PHY experiments and GPU profiling for Weaver, a system exploring opportunistic foundation model training on GPU-accelerated RAN infrastructure.
 - **RAN Emulation at Scale:** Contributed to system deployment, evaluation, and paper writing for EmuRAN, a cloud-based RAN emulation system supporting 10,000+ devices and base stations.

TECHNICAL SKILLS

- C/C++, Python, LLVM IR, Kubernetes, Open5GS, OAI, Linux, Docker

RESEARCH INTERESTS

- Networking and ML Systems, AI-RAN, Digital Twins, Distributed Edge Intelligence

SELECTED PUBLICATIONS

- **Morphling: Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
 - Developed a measurement-driven emulator for evaluating distributed ML training on heterogeneous edge devices at scale.
- **WASP: Foundation Model Training System for the Edge** Under Submission
 - Designed a foundation model training system for resource-constrained edge environments.
- **Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** Under Submission
 - Enabled opportunistic foundation model training on GPU-accelerated AI-RAN infrastructure.
- **EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** Under Submission
 - Built a cloud-based RAN emulation system capable of emulating 10,000+ mobile devices and base stations.
- **CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization** Under Submission
 - Developed an LLVM-based auto-quantization framework leveraging sensor specifications for embedded systems.

AWARDS

- **Edinburgh Award:** Recognition of professional growth and invaluable experiences gained as a Research Assistant.